

Nutrient Timing Recommendations Based on Population



Endurance Athlete

Common Training Patterns	Carbohydrates	Protein	Overall Priority
<i>Training:</i> Prolonged (1–4 hours) bouts of moderate intensity exercise each day causing significant disturbance in energy stores, fluid levels, and body temperature	<i>Pre:</i> 1–4 g/kg 1–3 hours prior to training or competition. Closer to competition, more aware individuals should be on gut comfort and hypoglycemia concerns <i>During:</i> 30–60 grams/hour of carbohydrate often as 8–12 fluid ounces of a 6–8% carbohydrate solution every 12–15 minutes during exercise <i>After:</i> Available recovery time is key driving factor. If 24 hours (or more), match carbohydrate intake with daily needs. If limited recovery, aggressive carbohydrate feedings should be considered.	<i>Pre:</i> Not a priority. Some protein or amino acid intake can occur and might help support protein synthesis, but focus should lie with carbohydrate delivery <i>During:</i> Similar to pre-exercise <i>After:</i> Proteins and amino acids should be incorporated in doses ranging from 8–12 grams of essential amino acids every 3–4 hours	• Daily carbohydrate needs will be much greater than many other types of athletes • Carbohydrate delivery and maintenance to fuel training and competition • Adequate protein during recovery to optimize repair and recovery

Nutrient Timing Recommendations Based on Population



Team-Sport, Strength and Power Athlete

Common Training Patterns	Carbohydrates	Protein	Overall Priority
<i>Training:</i> Depends on sport, but regular gym or field of play sessions ranging from 45 min to 2 hours. Mixture of sport technique, strength and power development, and anaerobic capacity optimization	<i>Pre:</i> Follow endurance athlete <i>During:</i> Follow endurance athlete. Although for some shorter sessions, carbohydrate intake might not be warranted. <i>After:</i> Similar to endurance athlete. Available recovery time is key driving factor. If 24 hours (or more), match carbohydrate intake with daily needs. If limited recovery, aggressive carbohydrate feedings should be considered.	<i>Pre:</i> Not a priority. Due to stronger desire to maximize strength and power development, some might be advocated towards ingestion of protein source delivering 8–12 grams of essential amino acids <i>During:</i> Maintain carbohydrate delivery. Few studies show positive changes in muscle protein synthesis with ingestion of EAAs or protein during resistance exercise sessions. <i>After:</i> Proteins and amino acids should be incorporated in doses ranging from 8–12 grams of essential amino acids every 3–4 hours	• Carbohydrate delivery to fuel intense training sessions • Adequate protein during recovery to aid in repair and recovery

Nutrient Timing Recommendations Based on Population



Recreational Athlete, Endurance

Common Training Patterns	Carbohydrates	Protein	Overall Priority
<i>Training:</i> Depending on seriousness, but sessions might be 4–5 times per week of 45–90 minutes	<i>Pre:</i> For longer sessions, follow pre-exercise strategies to maximize carbohydrate availability. Shorter sessions might not require any particular attention <i>During:</i> Depending on duration of session or amount of pre-workout fueling, carbohydrate intake during exercise might help support glucose levels and training intensity. For shorter sessions might not be needed <i>After:</i> Likely not as important as athlete should strive to consume meals and snacks to deliver carbohydrates to meet daily needs. If additional sessions are planned in short period, larger carbohydrate feedings shortly after session might be warranted.	<i>Pre:</i> Likely not a primary consideration <i>During:</i> Likely not a primary consideration <i>After:</i> Regular doses of high-quality sources of protein or amino acids delivering 8–12 grams of essential amino acids are warranted, but specific post-exercise timing is not required. Achieve daily protein needs.	Providing adequate energy, carbohydrate, and protein to meet daily needs

Nutrient Timing Recommendations Based on Population



Recreational Athlete, Strength and Power

Common Training Patterns	Carbohydrates	Protein	Overall Priority
<i>Training:</i> Depending on seriousness, but sessions might be 4–5 times per week of 45–60 minutes	<i>Pre:</i> Depending on intensity and duration of session, a small carbohydrate meal might help ensure adequate carbohydrate availability <i>During:</i> Likely not needed, add carbohydrates to fuel workout, maintain blood glucose, and vigor towards workout <i>After:</i> If total fueling picture suggests that post-exercise fueling is needed, then administer a post-workout meal. Might help reduce soreness and improve feelings throughout the day.	<i>Pre:</i> Likely not a primary consideration. No scientific reason to strongly consider. <i>During:</i> Likely not a primary consideration. No scientific reason to strongly consider. <i>After:</i> Regular doses of high-quality sources of protein or amino acids delivering 8–12 grams of essential amino acids are warranted, but specific post-exercise timing is not required. Achieve daily protein needs.	Providing adequate energy, carbohydrate, and protein to meet daily needs

Nutrient Timing Recommendations Based on Population



General Population – Health

Common Training Patterns	Carbohydrates	Protein	Overall Priority
<i>Training:</i> Depends on individual. 2–4 sessions per week for 30–50 minutes.	<i>Pre:</i> Likely not a primary consideration. No scientific reason to strongly consider. <i>During:</i> Likely not needed, add carbohydrates to fuel workout, maintain blood glucose, and vigor towards workout. <i>After:</i> If total fueling picture suggests that post-exercise fueling is needed, then administer a post-workout meal. Might help reduce soreness and improve feelings throughout the day	<i>Pre:</i> Likely not a primary consideration. No scientific reason to strongly consider. <i>During:</i> Likely not a primary consideration. No scientific reason to strongly consider. <i>After:</i> Regular doses of high-quality sources of protein or amino acids delivering 8–12 grams of essential amino acids are warranted, but specific post-exercise timing is not required. Achieve daily protein needs.	Providing adequate energy, carbohydrate, and protein to meet daily needs

Nutrient Timing Recommendations Based on Population



General Population – Physique

Common Training Patterns	Carbohydrates	Protein	Overall Priority
<i>Training:</i> Depends on individual. 2–4 sessions per week for 30–50 minutes with focus on energy restriction and fat loss.	<i>Pre:</i> Likely not a primary consideration. No scientific reason to strongly consider. <i>During:</i> Likely not needed, add carbohydrates to fuel workout, maintain blood glucose, and vigor towards workout <i>After:</i> If total fueling picture suggests that post-exercise fueling is needed, then administer a post-workout meal. Might help reduce soreness and improve feelings throughout the day.	<i>Pre:</i> Likely not a primary consideration. No scientific reason to strongly consider. <i>During:</i> Likely not a primary consideration. No scientific reason to strongly consider. <i>After:</i> Regular doses of high-quality sources of protein or amino acids delivering 8–12 grams of essential amino acids are warranted, but specific post-exercise timing is not required. Achieve daily protein needs.	Providing adequate energy, carbohydrate, and protein to meet daily needs